

Laboratory Work 1: ERD Diagram - International Airport

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1. Introduction

The goal of this lab is to design a simple database diagram (ERD) for an international airport. I will set up the main tables, connect them, and define the basic rules for how the airport database works.

2. Identified Entities

Entity	Purpose
Airport	Stores airport information: name, country, state, city, creation and update timestamps.
Airline	Stores airline company information including airline code, name, country, creation and update timestamps.
Flight	Stores scheduled and actual flight information, gates, airline, departure/arrival airports, and timestamps.
Passenger	Stores passenger personal and passport information.
Booking	Stores a passenger booking for a flight, its status, platform, ticket price, and timestamps.
Booking_Changes	Separate entity used to record changes made to a booking flight.
BoardingPass	Stores boarding pass information issued for a booking.
Baggage	Stores baggage registered during a booking, including weight and timestamps.
BaggageCheck	Stores the baggage checking result associated with a booking and passenger.
SecurityCheck	Stores airport security check information for a passenger.

3. Attributes, Keys and Data Types

Entity	Attribute	Type	Constraint	Description
Airport	airport_id	BIGINT	PK	Unique identifier.
Airport	airport_name	VARCHAR(150)	Required	Airport name.
Airport	country	VARCHAR(100)	Required	Country.
Airport	state	VARCHAR(100)	Optional	State/region.
Airport	city	VARCHAR(100)	Required	City.
Airport	created_at	TIMESTAMP	Required	Creation time.
Airport	updated_at	TIMESTAMP	Required	Last update time.
Airline	airline_id	BIGINT	PK	Unique identifier.
Airline	airline_code	VARCHAR(10)	Required / UQ	Airline code.
Airline	name	VARCHAR(150)	Required	Airline name.
Airline	country	VARCHAR(100)	Required	Country.
Airline	created_at	TIMESTAMP	Required	Creation time.
Airline	updated_at	TIMESTAMP	Required	Last update time.
Flight	flight_id	BIGINT	PK	Unique identifier.
Flight	departing_gate	VARCHAR(10)	Optional	Departure gate.
Flight	arriving_gate	VARCHAR(10)	Optional	Arrival gate.
Flight	airline_id	BIGINT	FK	References Airline.
Flight	departure_airport_id	BIGINT	FK	References Airport.
Flight	arrival_airport_id	BIGINT	FK	References Airport.
Flight	scheduled_departure_time	TIMESTAMP	Required	Scheduled departure time.
Flight	scheduled_arrival_time	TIMESTAMP	Required	Scheduled arrival time.
Flight	actual_departure_time	TIMESTAMP	Optional	Actual departure time.
Flight	actual_arrival_time	TIMESTAMP	Optional	Actual arrival time.
Flight	created_at / updated_at	TIMESTAMP	Required	Audit timestamps.

3. Attributes, Keys and Data Types - Continued

Entity	Attribute	Type	Constraint	Description
Passenger	passenger_id	BIGINT	PK	Unique identifier.
Passenger	first_name	VARCHAR(100)	Required	First name.
Passenger	last_name	VARCHAR(100)	Required	Last name.
Passenger	gender	VARCHAR(20)	Required	Gender.
Passenger	date_of_birth	DATE	Required	Date of birth.
Passenger	country_of_citizenship	VARCHAR(100)	Required	Citizenship country.
Passenger	country_of_residence	VARCHAR(100)	Required	Residence country.
Passenger	passport_number	VARCHAR(30)	Required / UQ	Passport number.
Passenger	created_at / updated_at	TIMESTAMP	Required	Profile timestamps.
Booking	booking_id	BIGINT	PK	Unique identifier.
Booking	flight_id	BIGINT	FK	References Flight.
Booking	passenger_id	BIGINT	FK	References Passenger.
Booking	status	VARCHAR(30)	Required	Booking status.
Booking	booking_platform	VARCHAR(50)	Required	Platform used for booking.
Booking	ticket_price	NUMERIC(10,2)	Required	Ticket price.
Booking	created_at / updated_at	TIMESTAMP	Required	Booking timestamps.
Booking_Changes	change_id	BIGINT	PK	Unique identifier for change record.
Booking_Changes	booking_id	BIGINT	FK	References Booking.
Booking_Changes	old_flight_id	BIGINT	FK (recommended)	Flight before change.
Booking_Changes	new_flight_id	BIGINT	FK (recommended)	Flight after change.
Booking_Changes	change_reason	VARCHAR(255)	Optional	Reason for change.
Booking_Changes	changed_at	TIMESTAMP	Required	Change time.

3. Attributes, Keys and Data Types - Continued

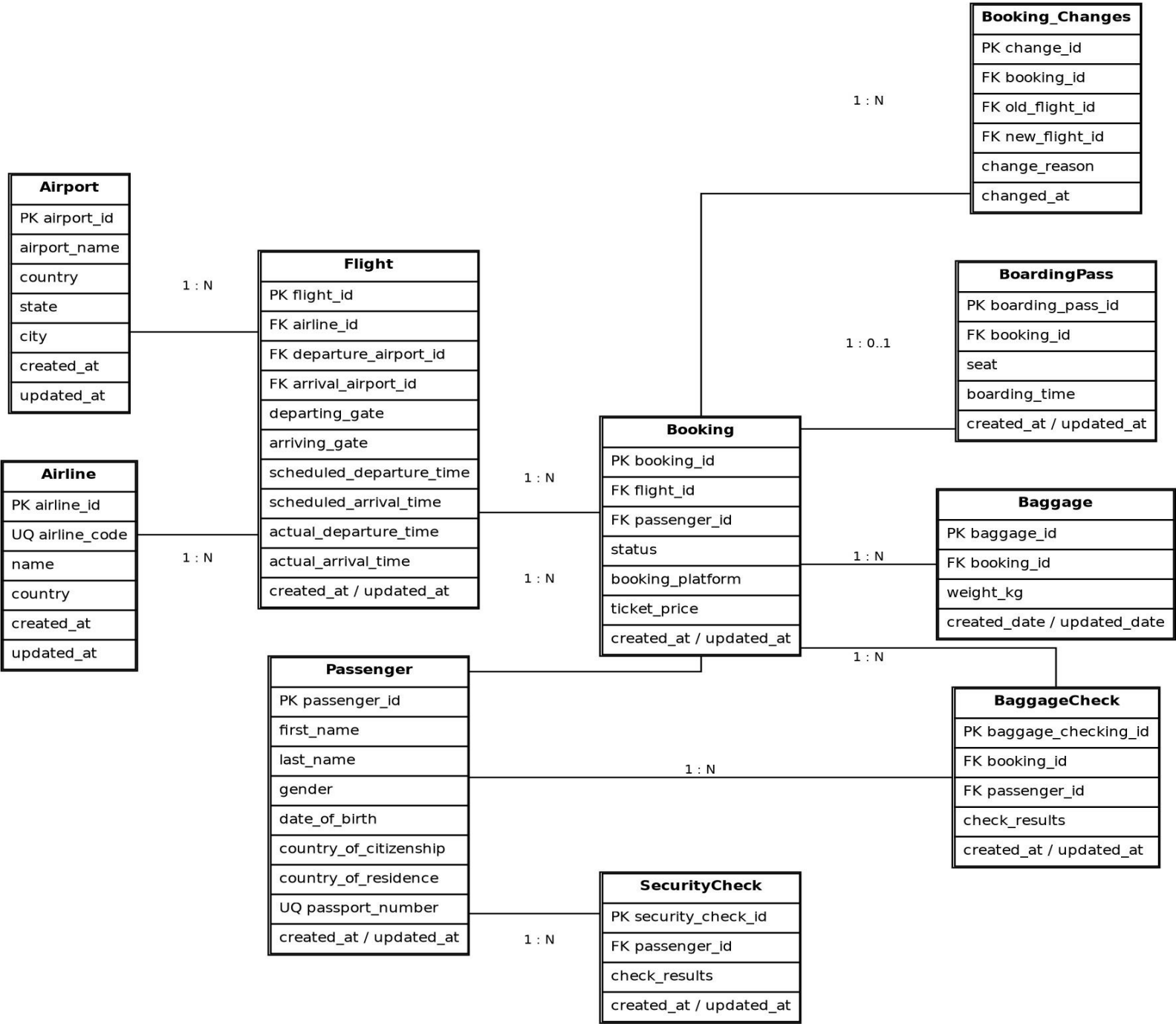
Entity	Attribute	Type	Constraint	Description
BoardingPass	boarding_pass_id	BIGINT	PK	Unique identifier.
BoardingPass	seat	VARCHAR(10)	Required	Assigned seat.
BoardingPass	boarding_time	TIMESTAMP	Required	Boarding time.
BoardingPass	created_at / updated_at	TIMESTAMP	Required	Pass timestamps.
BoardingPass	booking_id	BIGINT	FK	References Booking.
Baggage	baggage_id	BIGINT	PK	Unique identifier.
Baggage	weight_kg	NUMERIC(6,2)	Required	Weight in kilograms.
Baggage	created_date / updated_date	TIMESTAMP	Required	Baggage timestamps.
Baggage	booking_id	BIGINT	FK	References Booking.
BaggageCheck	baggage_checking_id	BIGINT	PK	Unique identifier.
BaggageCheck	check_results	VARCHAR(30)	Required	Check result.
BaggageCheck	created_at / updated_at	TIMESTAMP	Required	Check timestamps.
BaggageCheck	booking_id	BIGINT	FK	References Booking.
BaggageCheck	passenger_id	BIGINT	FK	References Passenger.
SecurityCheck	security_check_id	BIGINT	PK	Unique identifier.
SecurityCheck	check_results	VARCHAR(30)	Required	Security check result.
SecurityCheck	created_at / updated_at	TIMESTAMP	Required	Check timestamps.
SecurityCheck	passenger_id	BIGINT	FK	References Passenger.

4. Relationships and Cardinalities

Relationship	Cardinality	Simple explanation
Airport - Flight (departure)	1 : N	One airport can be the departure airport for many flights; each flight has one departure airport.
Airport - Flight (arrival)	1 : N	One airport can be the arrival airport for many flights; each flight has one arrival airport.
Airline - Flight	1 : N	One airline operates many flights; each flight belongs to one airline.
Passenger - Booking	1 : N	One passenger can make many bookings; each booking references one passenger.
Flight - Booking	1 : N	One flight can have many bookings; each booking references one flight.
Booking - BoardingPass	1 : 0..1	A booking may have zero or one boarding pass; each boarding pass belongs to one booking.
Booking - Baggage	1 : N	One booking can contain multiple baggage records; each baggage record belongs to one booking.
Booking - BaggageCheck	1 : N	A booking can have multiple baggage-check records; each record references one booking.
Passenger - BaggageCheck	1 : N	A passenger can have multiple baggage-check records; each record references one passenger.
Passenger - SecurityCheck	1 : N	A passenger can have multiple security checks; each security check belongs to one passenger.
Booking - Booking_Changes	1 : N	A booking can have multiple recorded flight changes; each change record belongs to one booking.

5. Entity-Relationship Diagram

PK means Primary Key, FK means Foreign Key, and UQ means Unique constraint.



6. ERD Legend / Notation

Notation	Meaning
PK	Primary Key - uniquely identifies a row.
FK	Foreign Key - references a primary key in another entity.
UQ	Unique constraint - prevents duplicate values.
1 : N	One-to-Many relationship.
1 : 0..1	One-to-zero-or-one relationship.
Required	Attribute is expected to contain a value.
Optional / NULL	Value may be absent, especially for actual flight times before a flight operates

7. Conclusion

Overall, this diagram shows how the airport database works. Primary and foreign keys make sure all the data stays accurate and connected. It splits everything into clean tables like passengers, baggage, and flights, making it easy to implement in PostgreSQL.